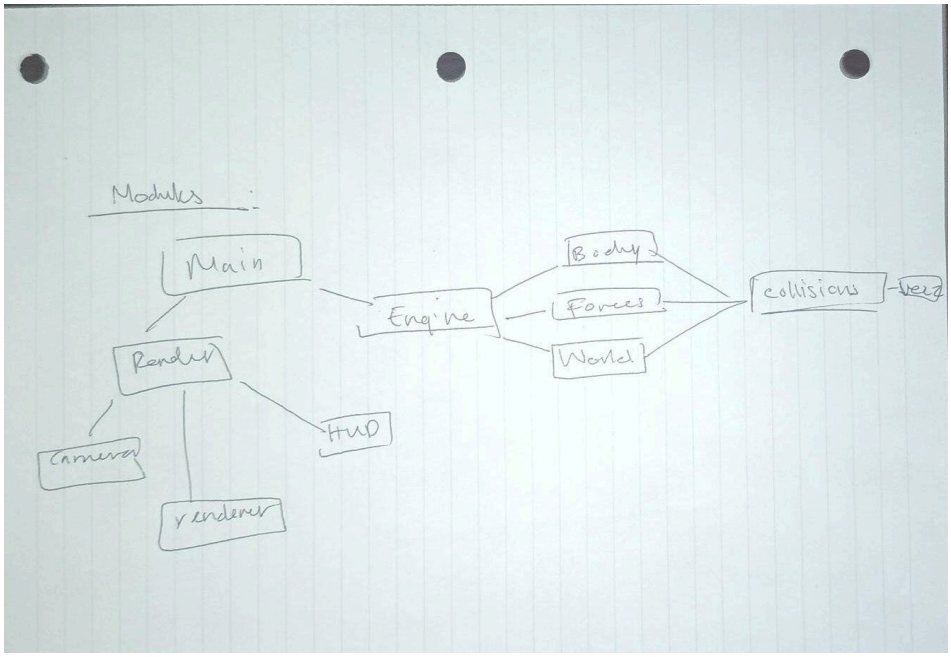
1. Photo of module dependency diagram

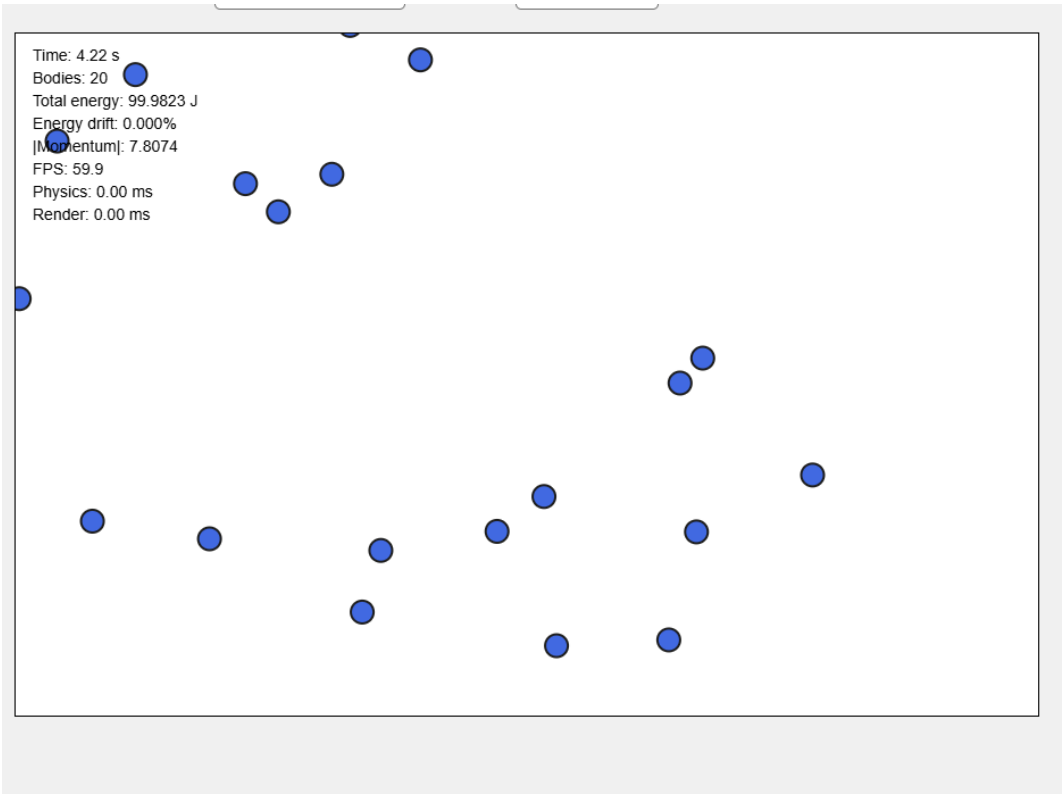


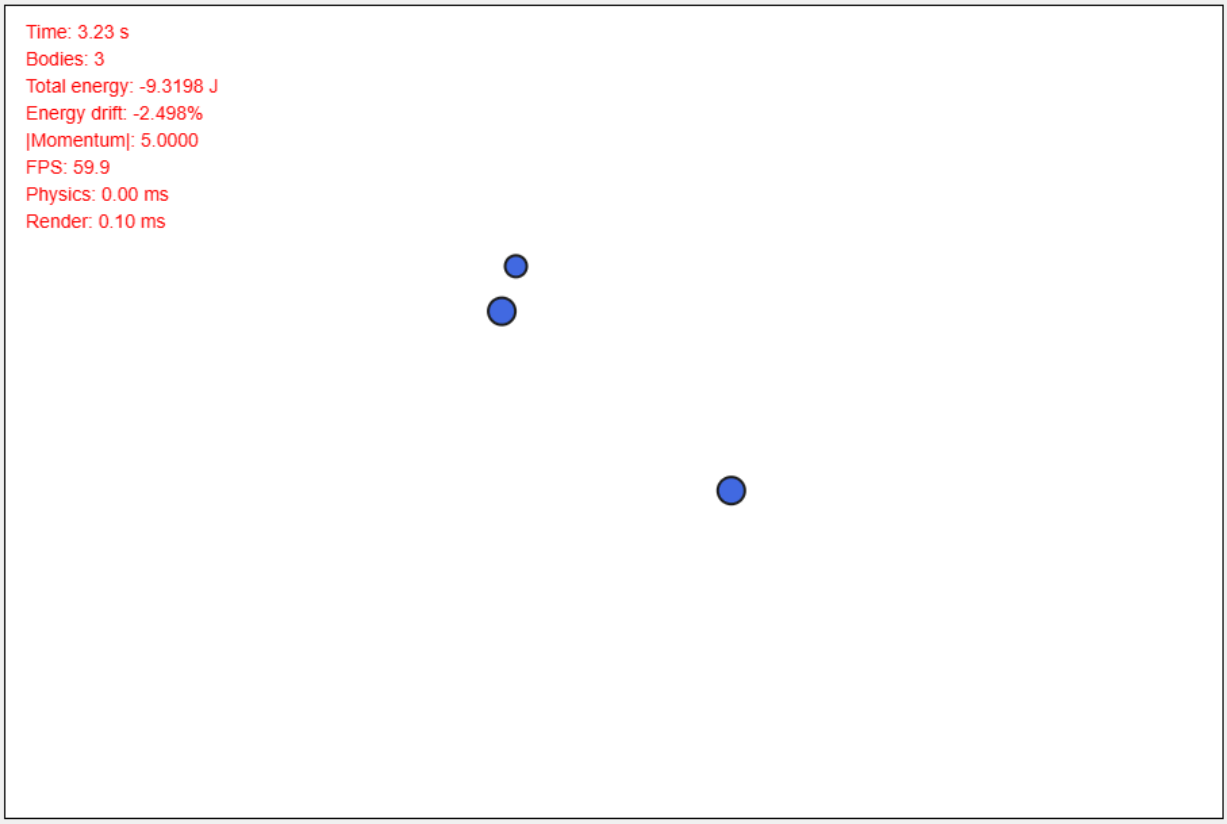
2. Jitterbug: This is where the circle collision comes into play. The algorithm checks the V of the overlapping balls. The impulse condition gets overlooked or skipped if the velocity is positive.
   You can see the demonstration when:
   If (v > 0) continue;

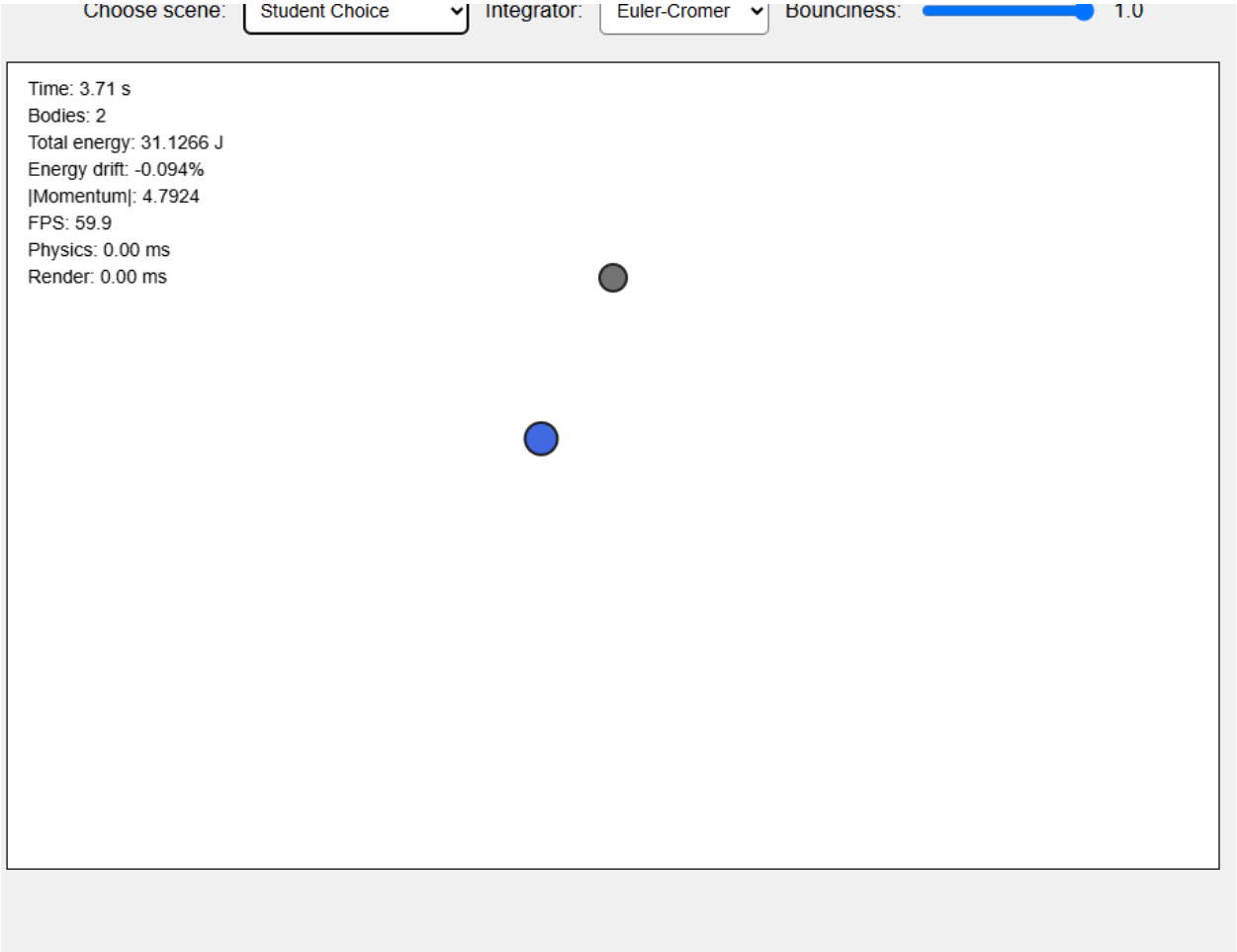
   When I remove this, you can see extra bouncing and changes in v to high or low.

3. Test results:
   a. Condition 1: E collision: no loss of KE, p is measured
   b. Condition 2: IE collision: KE is loss from balls per collision, e and p get measured
   c. Condition 3:E stays constant, e drift gets measured

4. Scenes: 20 bodies for Billiards, N-body, and Student


Time: 4.22 s
Bodies: 20
Total energy: 99.9823 J
Energy drift: 0.000%
|Momentum|: 7.8074
FPS: 59.9
Physics: 0.00 ms
Render: 0.00 ms


Time: 3.23 s
Bodies: 3
Total energy: -9.3198 J
Energy drift: -2.498%
|Momentum|: 5.0000
FPS: 59.9
Physics: 0.00 ms
Render: 0.10 ms



5. ***Asked Chatgpt for chart template, but data is from simulation***

Bodies	Physics (ms)	Rendering (ms)	FPS
20	0.10	0.1	59.9
50	0.10	0.10	60
100	0.20	0.10	59-60
200	0.6	0.2	59.9

Bottleneck Analysis:
A bottleneck in physics is the point at which our simulations slow down or crowd. Here, I think the bottleneck occurs during eany nested loop since the simulation must push through each on individually, especially when we increase the number of balls.

6. To be honest, I reused the exact same code. The only thing we need to change in this engine is the variable properties, bodies, forces.